

Competitive Programming

Section 61: ICPC Readiness – Offline Problem Sheet

Competitive Programming Course

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1 Offline use

This is an original concise sheet for an ordered ICPC-style ladder. Each link leads to the authoritative external statement and judge. Work from this sheet offline, but read the official statement before submitting: it controls any detail not repeated here. Record a model and a hostile test before reading the editorial.

Each rung also has an original, locally judgeable companion package under `problems/`; see [the companion index](#). The companion is a separate exercise for the same course skill, with skeletons, references, and tests. It is intentionally not a copy of the linked Kattis statement.

2 Round 1: reliable opening solves

2.1 1. ACM Contest Scoring

Official: <https://open.kattis.com/problems/acm>

Replay one team's submissions. A wrong answer before a problem's first acceptance increases that problem's wrong-attempt count; a later acceptance adds its minute plus the stated penalty per earlier wrong answer. Ignore all records after the contest-end marker. Print solved count and penalty.

2.2 2. Booking a Room

Official: <https://open.kattis.com/problems/bookingaroom>

Given room numbers already reserved in a hotel numbered from 1 through R , print the smallest available room number, or state that every room is booked. The input order of reservations is arbitrary.

2.3 3. Sort of Sorting

Official: <https://open.kattis.com/problems/sortofsorting>

For each dataset of uppercase names, sort the names stably by their first two characters. Preserve the required blank-line convention between datasets.

2.4 4. ICPC Team Selection

Official: <https://open.kattis.com/problems/icpcteamsselection>

From $3n$ student skill values, repeatedly form teams of three. The score of a team is its median skill. Maximize the sum of all team scores.

2.5 5. Guess the Data Structure

Official: <https://open.kattis.com/problems/guessthedatastructure>

Observe insert and remove-value operations. Decide whether the observations are consistent with a stack, queue, priority queue, more than one of them, or none.

3 Round 2: regional implementation and modeling

3.1 6. Assigning Workstations

Official: <https://open.kattis.com/problems/workstations>

Researchers arrive at known times and use a workstation for known durations. After use, a workstation remains unlocked for a fixed timeout. Minimize the number of unlock operations by reusing a suitable unlocked machine.

3.2 7. Teque

Official: <https://open.kattis.com/problems/teque>

Maintain a sequence under push-front, push-back, push-middle, and get-index operations. There can be many operations, so inserting into a normal array is too slow.

3.3 8. Almost Union-Find

Official: <https://open.kattis.com/problems/almostunionfind>

Maintain disjoint sets under union, moving one element from its current set to another set, and queries for a set's size and sum of element labels.

3.4 9. Single Source Shortest Path

Official: <https://open.kattis.com/problems/shortestpath1>

For repeated directed weighted graphs with nonnegative weights, answer shortest distance queries from a specified source. Print `Impossible` for unreachable vertices.

3.5 10. Get Shorty

Official: <https://open.kattis.com/problems/getshorty>

Edges multiply a traveller's height by a factor between zero and one. Find a path from the first vertex to the last that maximizes the final height factor.

4 Round 3: advanced regional tools

4.1 11. Coast Length

Official: <https://open.kattis.com/problems/coast>

Given a grid of land and water, count only land edges exposed to the outside ocean. Water enclosed by land is a lake and does not contribute coastline.

4.2 12. All Pairs Shortest Path

Official: <https://open.kattis.com/problems/allpairspath>

Answer distance queries in directed graphs that may contain negative edges. Distinguish unreachable pairs from pairs whose shortest walk is unbounded below because a reachable negative cycle can still reach the destination.

4.3 13. Arbitrage

Official: <https://open.kattis.com/problems/arbitrage>

Given exchange rates between currencies, determine whether a sequence of trades can strictly increase the amount of some starting currency.

4.4 14. Knapsack

Official: <https://open.kattis.com/problems/knapsack>

For each capacity and item list, maximize total value with 0/1 choices and output one set of selected item indices achieving that value.

4.5 15. Clock Pictures

Official: <https://open.kattis.com/problems/clockpictures>

Two circular sets of clock-hand angles represent the same picture when one can be rotated to match the other. Decide whether such a rotation exists.

5 Round 4: finalist-style reductions

5.1 16. String Matching

Official: <https://open.kattis.com/problems/stringmatching>

For each pattern and text pair, print all starting positions where the pattern occurs, including overlapping occurrences.

5.2 17. Max Flow

Official: <https://open.kattis.com/problems/maxflow>

Given a directed capacitated network, compute a maximum flow from source to sink and output a valid positive-flow decomposition on original edges.

5.3 18. Min Cut

Official: <https://open.kattis.com/problems/mincut>

For a directed capacitated network, output the source-side vertex set of a minimum cut after computing a maximum flow.

5.4 19. Distributing Ballot Boxes

Official: <https://open.kattis.com/problems/distributingballotboxes>

Distribute a fixed number of ballot boxes among cities, giving every city at least one. Minimize the largest population served by one box.

5.5 20. Catering

Official: <https://open.kattis.com/problems/catering>

Events require a known number of meals. Meals can be prepared, reused after an event at a cleaning cost, or newly purchased; travel/reuse costs are given. Find the minimum total cost over the complete event schedule.

6 Attempt record

For every task, write this before opening the editorial:

```
model: _____ invariant: _____  
first hostile test: _____ exact blocker: _____
```